

Dynamic Narrative Personalization in Games Using AI

BSc (Hons) Artificial Intelligence and Data Science

CM4609 — Individual Research Project

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The Problem

Current game narrative systems are **static**: they do not adapt their tone, urgency, or content to the psychological state and demonstrated preferences of an individual player in real time.

What the player gets

- The same village elder warning regardless of playstyle
- The same quest text whether exploring or rushing
- No narrative response to struggle, boredom, or curiosity

What the player needs

- A story that notices when challenge exceeds skill
- Narrative tone that matches gameplay intensity
- Content that responds to *how* they play, not just *what* they choose

The cost: reduced engagement in the short term; a missed opportunity to use narrative as a lever for nudging players toward optimal experience.

Literature and Research Gap

Prior work clusters into three categories, and each leaves a gap:

Approach	Example	What's missing
Authored branching	Baldur's Gate 3, Disco Elysium	Responds to explicit choices, not behavioural signals
Procedural generation	Tracery, Expressionist	No player model; no psychological sensing
LLM-based narrative	PANGeA (AAAI 2024)	Static prompts; no RL adaptation; no episodic memory

Six research domains converge in this work:

Player Modelling (Hexad) + **Flow Theory** (Csikszentmihalyi) + **RL Adaptation** (PPO) + **Narrative Personalization** + **LLMs for Games** (RAG) + **Episodic Memory** (EM-LLM)

The gap is integration, not invention. Each technique exists in isolation. No prior work has wired them into a single closed loop.

Research Questions

RQ1: Can a closed-loop AI system driven by continuous Hexad profiling and Flow-state detection produce narrative that players perceive as **more personalized** than static narrative?

RQ2: Does PPO-driven narrative adaptation lead to measurably **higher Flow experience** (GEQ Flow subscale) compared with a control condition?

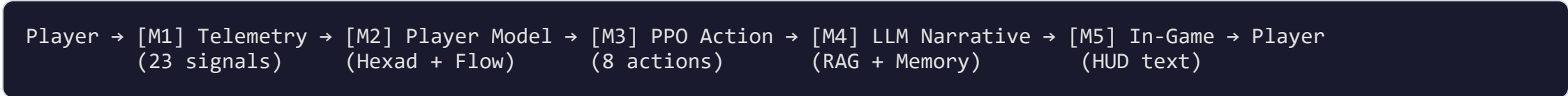
RQ3: How do players **perceive and experience** dynamically personalized narrative in terms of engagement and immersion?

Study design

Between-subjects, N = 50, Welch's *t*-test, Bonferroni correction ($\alpha_{\text{adj}} = .0167$)

System Architecture: Five-Module Closed Loop

The pipeline cycles every 5 seconds from player behaviour back to in-game delivery:



Module	Key Component	Technology
M1 Telemetry	PlayerDataLogger	Unity C#, WebSocket, 5s windows
M2 Player Model	FeatureExtractor → HexadProfiler → FlowClassifier	Python, rule-based decision tree
M3 Adaptation	NarrativeAgent (PPO)	Stable-Baselines3, Gymnasium MDP
M4 Generation	LoreRetriever → EpisodicMemory → PromptBuilder → OllamaClient	sentence-transformers, Ollama
M5 Injection	NarrativeManager → ContentInjector	Unity C#, UnityEvents

Platform split: Unity client (M1 + M5) ↔ FastAPI backend (M2 + M3 + M4)

Player Modelling: Hexad + Flow

Hexad from Telemetry (no questionnaire)

23 raw signals → 11 features → 6 continuous
Hexad scores

Dimension	Behavioural Proxies
Achiever	Objective rate, KD ratio
Explorer	Map coverage, lore engagement
Socialiser	Voluntary NPC talks, dialogue
Free Spirit	Exploration off critical path
Disruptor	Kills, deaths, damage dealt
Philanthropist	Dialogue + lore engagement

Primary novelty claim: first system to derive Hexad profiles entirely from behavioural

Flow Classification

Rule-based decision tree on **challenge-skill ratio**:

$$\begin{aligned} \text{challenge} &= 0.50 \cdot \text{dmg} + 0.30 \cdot (1 - \text{obj}) + 0.20 \cdot \text{deaths} \\ \text{skill} &= 0.35 \cdot \text{kd} + 0.30 \cdot \text{obj} + 0.20 \cdot \text{acc} + 0.15 \cdot (1 - \text{idle}) \\ \text{ratio} &= \text{challenge} / \text{skill} \end{aligned}$$

State	Condition
APATHY	idle > 0.40 AND dial < 0.30 AND ratio < 0.60
BOREDOM	ratio < 0.75
ANXIETY	ratio > 1.30
FLOW	$0.75 \leq \text{ratio} \leq 1.30$

Window: last 6 batches (~30s) for Flow; full session for Hexad

RL-Based Adaptation: PPO for Narrative

MDP formulation

- **State:** (FlowState, HexadProfile) = 4 discrete × 6 continuous dimensions
- **Actions:** 8 narrative modulations:

Action	Purpose
RAISE_STAKES	Increase tension
LOWER_STAKES	Reduce pressure
ADD_MYSTERY	Spark curiosity
ADD_HUMOR	Lighten tone

Action	Purpose
PROVIDE_GUIDANCE	Help struggling players
INCREASE_URGENCY	Accelerate pace
LORE_REWARD	Reward explorers
NO_CHANGE	Maintain current

Flow-based reward function (graduated penalties)

FLOW	BOREDOM	ANXIETY	APATHY
+1.0	-0.3	-0.5	-0.8

Narrative Generation: RAG + Episodic Memory

RAG pipeline

- **Lore corpus:** ~30 paragraphs (world, characters, quests)
- **Embedding:** all-MiniLM-L6-v2 (384-dim, L2-normalised)
- **Retrieval:** in-memory numpy cosine similarity; top-3 chunks
- **Dynamic:** same location + different Flow state = different lore retrieved

LLM

- **Phi-3.5 Mini** (3.8B params) via Ollama
- Runs entirely locally: no cloud, no API

Episodic session memory

Inspired by EM-LLM (Fountas et al., 2024):

- **Importance-weighted** events (score 0-1)
- Capped at 10 events; evicts lowest importance
- Top 5 by importance injected into prompt
- Enables cross-event references and character continuity

Prompt structure

```
SYSTEM: Role + tone (from Flow state)
        + output constraints
USER:   Player state + Hexad profile
        + RAG lore context (3 passages)
        + Memory context (5 events)
```


Implementation

Backend (Python)

- **FastAPI** with 4 routers: telemetry (WS + HTTP), player-model, narrative
- **SQLite** for telemetry persistence
- **Stable-Baselines3** PPO with Gymnasium MDP env
- **sentence-transformers** for RAG embeddings
- **Ollama HTTP client** for local LLM inference
- **8 test files** (pytest) covering all layers

Unity Plugin (C# / Unity 6)

- UPM package: `com.research.narrative-plugin`
- `PlayerDataLogger` : 23 signals every 5s via WebSocket
- `NarrativeManager` : polls `/api/narrative`
- `ContentInjector` : fires `OnNarrativeReady` UnityEvent
- **Testbed**: 3D RPG on AnyRPG Core (Eryndal world)
- TMLPro HUD panel with auto-dismiss and fade

Evaluation Design

Between-subjects experiment

- **N = 50** (25 Experimental, 25 Control)
- **30-minute** single session per participant
- **Experimental:** full 5-module adaptive system
- **Control:** same game, same triggers, static `ScriptableObject` narrative
- Everything else identical: environment, objectives, NPCs, UI, session length

Instruments

Scale	What it measures
GEO Flow ($\alpha = .79$)	Self-reported Flow experience

Qualitative component

- **20 semi-structured interviews** (10 per condition)
- Reflexive Thematic Analysis (Braun & Clarke, 2022)
- 47 initial codes → 4 final themes

Statistical approach

- **Welch's *t*-test** (robust to unequal variance)
- **Bonferroni correction:** $\alpha_{\text{adj}} = .0167$ for 3 hypotheses
- **Cohen's *d*** effect sizes with 95% CIs

Results: Quantitative Findings

d = 0.84**

NPS-3 Personalization
p = .005 (survives Bonferroni)

d = 0.65*

GEQ Flow
p = .027 (misses Bonferroni)

d = 0.38

GEQ Immersion
p = .187 (n.s.)

Measure	Exp. M (SD)	Ctrl. M (SD)	Cohen's d	p	Bonferroni?
NPS-3	4.92 (1.18)	3.88 (1.31)	0.84	.005	Survives
GEQ Flow	2.78 (0.62)	2.36 (0.68)	0.65	.027	Misses
GEQ Immersion	2.64 (0.71)	2.38 (0.66)	0.38	.187	No

Results: Qualitative Themes

20 interviews analysed via Reflexive Thematic Analysis (Braun & Clarke, 2022):

Theme 1: "The story pays attention" (8/10 Exp.)

Ambient sense that the narrative noticed and responded to *how* they played. Metaphor of being "seen" by the game. *"It felt like the game knew I was exploring. The Chronicler started telling me things that matched what I was actually doing."* — P07

Theme 2: "Tone matching" (6/10 Exp.)

Urgent dialogue during combat; lore-rich content during exploration. Players noticed and valued the alignment. *"When I was struggling with the enemies, the dialogue got more urgent, more helpful. That felt right."* — P12

Theme 3: "Generic narrative as wallpaper" (7/10 Ctrl.)

Static narrative perceived as irrelevant, not bad. Gradual disengagement. Corroborates 58% dialogue read ratio.

Theme 4: "Occasional dissonance" (4/10 Exp.)

Tone mismatch during rapid state transitions. Root cause: 5-second telemetry lag.

Discussion and Limitations

What the pattern means

Strong personalization → **moderate Flow** → **weak immersion** is internally coherent:

- The system directly manipulates narrative content (NPS-3 captures this)
- This manipulation has a downstream effect on Flow (the PPO optimisation target)
- Immersion depends on factors the system does not control (visuals, audio, polish)

Honest limitations

- **PPO trained in simulation**, not on real player data. Markovian assumption almost certainly violated in practice
- **Hexad profiling is rule-based**; no ground-truth questionnaire validation. Cannot distinguish intentional from
- **Single 30-minute session**; cumulative effects invisible
- **Prototype testbed**, not a commercial game; limits ecological validity
- **N = 49**; powered for medium effects, blind to small ones

Contributions, Ethics, and Future Work

Six original contributions

1. **Telemetry-derived Hexad profiles** — no questionnaire; within-session updating
2. **Dynamic RAG-grounded narrative** — lore context selected at runtime by player state
3. **Episodic session memory** for real-time game narrative (first application)
4. **PPO for narrative action selection** — reframes RL from mechanical DDA to story
5. **Flow-based MDP reward** — principled theoretical grounding

SPER Considerations

Social: Player profiling raises questions about behavioural surveillance in games. Flow-sustaining systems nudge psychological states; this is a mild form of manipulation, acknowledged openly

Professional: System runs entirely locally. Open-weight models, no commercial API dependency. All code, instruments, and analysis scripts included in submission

Ethical: Informed consent obtained; participants debriefed post-session. SPER form approved (14 Nov 2025; supervisor affirmed

Thank You — Questions?

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Key numbers

N = 49	Between-subjects study
d = 0.84	Personalization (NPS-3)
d = 0.65	Flow (GEQ)
d = 0.87	Dialogue read ratio
5 modules	Closed-loop pipeline
8 actions	PPO narrative selection
5-second	Telemetry cycle
23 signals	Per telemetry window

One-sentence summary